

WANG XIANGBO (王相博)

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🎓 Education

Undergraduate in Communication Engineering

2023 – 2027

Hangzhou Dianzi University

Top 10% in major | Coursework: Algorithms, Digital Signal Processing, Deep Learning, Linear Algebra, Probability & Statistics, Neural Networks, Digital Communications

Summer School

Summer 2024

The University of Sydney, Australia Fundamentals of Deep Learning, Academic English; earned **Excellent** in both final assessments. Short-term exchange; exposure to international academic environment.

High School Diploma

2020 – 2023

China

Chinese Biology Olympiad (Provincial 3rd Prize)

Chinese Physics Olympiad (Provincial 3rd Prize)

⚙️ Research Experience

Research Assistant

2024 – Present

I IPL (Intelligent Information Processing Lab), HDU

- Advised by Wenbin Jiang, Jin Wang. Research on deep learning-based neural audio codec.
- Designed SwitchCodec with Residual Experts Vector Quantization (REVQ); proposed dynamic routing for variable-bitrate coding without retraining.
- Achieved 2.87 PESQ @ 2.6kbps; SOTA on LibriTTS. Paper accepted at ICASSP 2026 (first-author).
- Implemented models in PyTorch; conducted ablation studies and subjective listening tests.
- Built and released demo website for audio samples (speech & music).

Research Assistant

2025 – Present

MILab (Media Intelligence Lab), HDU

- Advised by Zhenzhong Kuang. Research on 3D generation and super-resolution.
- Co-first (2nd) on SR3R: feed-forward Gaussian Splatting for super-resolution 3D reconstruction.
- Explored efficient rendering pipelines and multi-view 3D recovery; paper accepted by CVPR 2026.
- Contributed to architecture design and experimental validation on benchmark datasets.

📄 Publications

- [1] X. Wang, W. Jiang, J. Wang, Y. You, S. Fang, F. Wen. "Switchcodec: Adaptive Residual-Expert Sparse Quantization for High-Fidelity Neural Audio Coding." *ICASSP*, 2026. [arXiv](#) | [Demo](#)
- [2] X. Feng, X. Wang, T. Zhong, C. Wang, Y. Zhao, T. Xu, Z. Kuang, F. Qin, X. Yin, Y. Zhu. "SR3R: Rethinking Super-Resolution 3D Reconstruction With Feed-Forward Gaussian Splatting." *CVPR*, 2026. (Accepted, co-first 2nd)

🏆 Awards & Honors

- **University Scholarship** (academic excellence) | **Top 10%** in major ranking
- **Chinese Biology Olympiad** (Provincial 3rd Prize)
- **Chinese Physics Olympiad** (Provincial 3rd Prize)
- ICASSP 2026 paper accepted (first-author)
- CVPR 2026 paper accepted (co-first, 2nd)

🔧 Technical Skills

Programming: Python (proficient), MATLAB | **Frameworks:** PyTorch, TensorFlow | **Tools:** CUDA, Docker, Git, LaTeX, W&B, TensorBoard

Environment: Linux (server, multi-GPU training) | **Languages:** Mandarin (Native), English (CET-4; IELTS in prep.) | *Research:* Audio coding, 3DGS, super-resolution, MLLM

🌟 Extracurricular & Interests

Academic: Member, HDU CS Association; active in lab seminars and reading groups.

Personal: Basketball (competitive, shooting guard); Saxophone (jazz & classical); Weiqi (1-dan 2016 → 5-dan). Collaboration, discipline, strategic thinking.